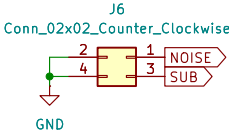
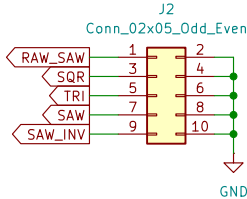
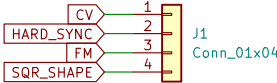
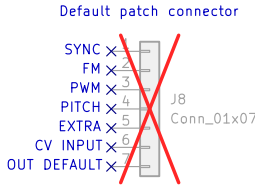
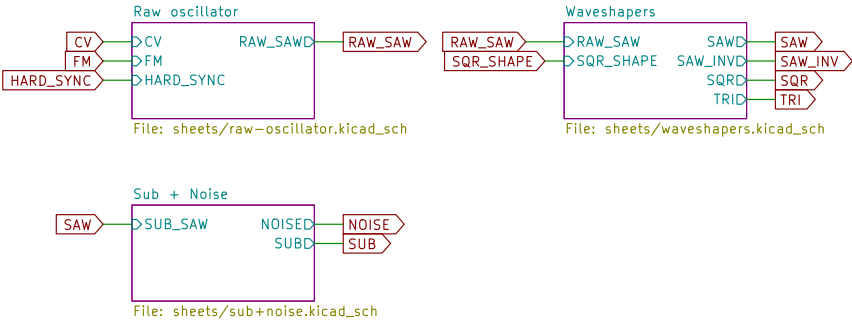


Raw oscillator & wave shaping

I/O



Shielding pairs
DO NOT GROUND ON UI BOARD!

Power



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Shmørergh		
Sheet: /		
File: vco-core.kicad_sch		
Title: Moduleur VCO core		
Size: A4	Date: 2026-05-24	Rev: 3
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CV inputs can be anything summed passively with resistors.
ALL INPUTS MUST HAVE AN INPUT RESISTOR
 For 1V/oct, use 100k

KEYBOARD PITCH X
 TUNE POT X
 FINETUNE POT X

#R1
100k
 #R2
100k
 #R3
100k

CVD

+12V
 R_BIAS1
100k (prec)
 -12V

Set output to C0 with BIAS when no pitch CV is applied.

FMD

+12V
 U1C
TL072
 U2C
TL072
 C1
100n
 C2
100n
 C3
100n
 C4
100n
 C5
100n
 C6
100n
 GND -12V

The following temperature compensation setups are supported in the schematics.
 V1 uses a 1206 SMD component (PTS120602B100RP100). V2 uses a THT NTC component (NTCLE203E3103FB0). Both works pretty much the same, they're here as a fallback for potential components shortage.

X: don't place component
 Short: short component
 Use multi-turn precision trimmers.

Temperature compenstion options:

	R_FB	R_W1	R_D1	TH1	R_D2	R_D3	TH2	R_W2
V1:	91k	20k	Short X		6.2k(50ppm)		4.22(50ppm)	PTS120602B100RP100 X
V2:	22k	Short 11k		NTCLE203E3103FB0	27k		2.7k	X 500ohm

Tuning:
 0. Connect a tuner and a keyboard. Play the C1 note.
 1. Turn Hi-Freq all the way down so there's no compensation: Play a very high pitch note and start turning the trimpot so that the pitch is going down. Once it doesn't go lower then you "turned off" high freq compensation.
 2. Use R_BIAS to set the frequency to C1 (32.703Hz)
 3. Play C4 and use R_BIAS to set the frequency to C4 (261.63Hz)
 4. Play C1 and use R_WIDTH to tune it to 32.703Hz LOOP 3 + 4
 5. When C1 and C4 are in tune, play higher octave C's and use R_HIFREQ to adjust tuning for them.
 If needed, repeat steps 3 and 4 for other note pairs.

ONLY ONE OF THESE J111's SHOULD BE SOLDERED.
 Q3 is a THT footprint for J111 to provide alternative to SMD.

+12V

R10
1M

TP1
-18mV/V_{REF}

Q1B
BC847BS

Q1A

R8
See t.comp. table

TP3
RAW_SAW

RAW_SAW

R11
1.5k

TP2
HiFreq

R_HIFREQ1
2k (prec)

-5V

Q2
MMBT3906

HI_FREQ_COMP

R12
510

R13
1M

R14
10k

R15
100k

-12V

Q5
MMBT5551

TP4
VCOVREF

TP5
Reset OUT

TP6
Reset OUT

TP7
Reset OUT

TP8
Reset OUT

TP9
Reset OUT

TP10
Reset OUT

TP11
Reset OUT

TP12
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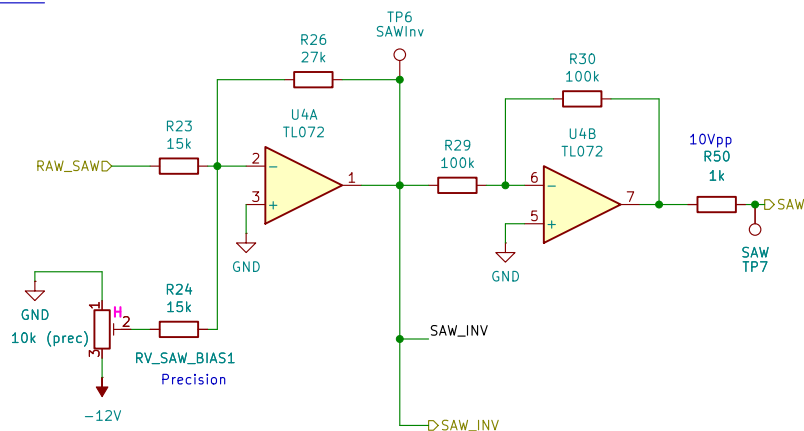
TP219
Reset OUT

TP220
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TP221
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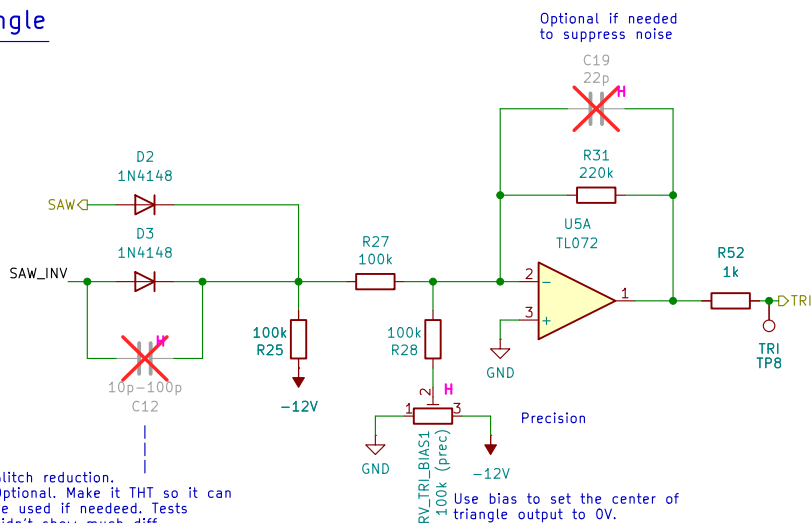
TP222
Reset

Sawtooth



Use BIAS to set the sawtooth output exactly to oscillate around 0V. This is critical for a nice triangle wave.

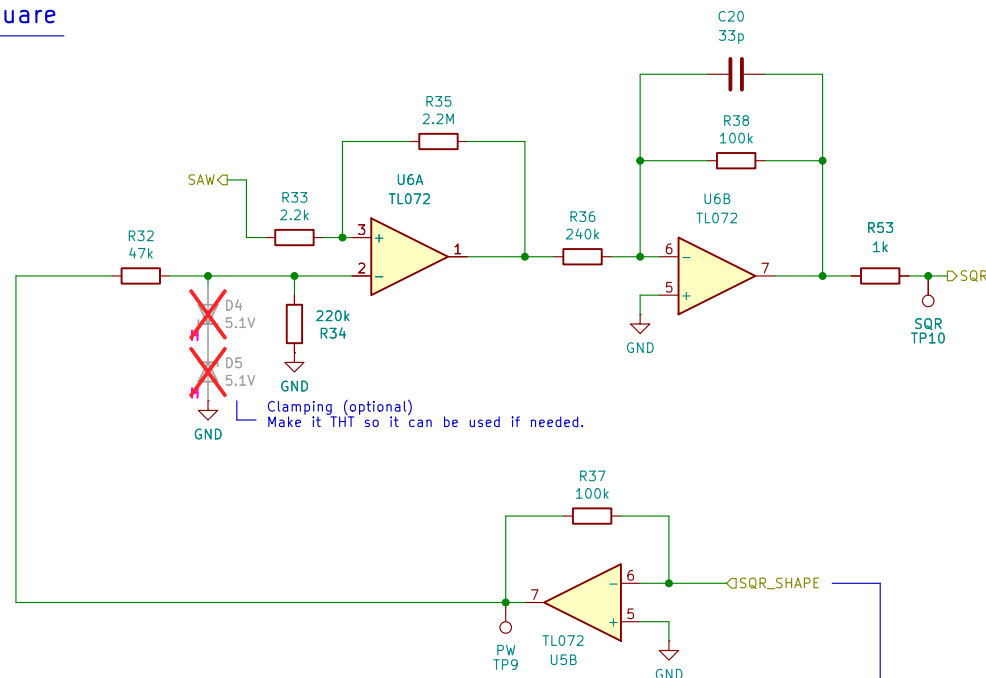
Triangle



Glitch reduction. Optional. Make it THT so it can be used if needed. Tests didn't show much diff.

Use bias to set the center of triangle output to 0V.

Square



Clamping (optional)
Make it THT so it can be used if needed.

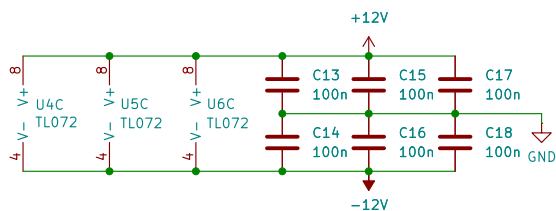
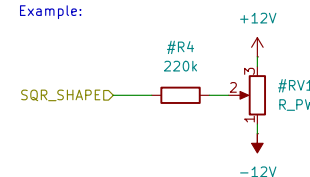
Square pulse width

Connect any number of CV inputs through input resistors to set the pulse width with a CV on SQR_SHAPE. Use the following CV values:

0V/GND: 50%
-5.5V: 5%
+5.5V: 95%

Set the input resistors so that the CV mixer's output value is between -/+5.5.

Example:



Shmøergh

Sheet: /Waveshapers/
File: waveshapers.kicad_sch

Title: **Moduleur VCO core**

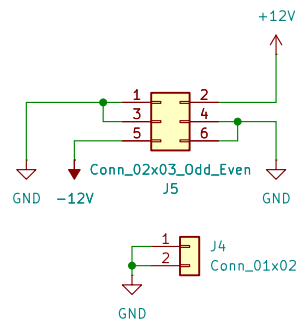
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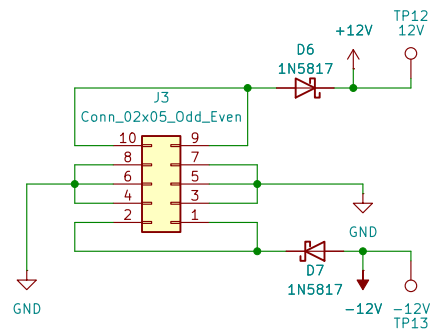
Rev: 3

Id: 3/5

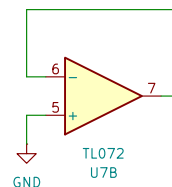
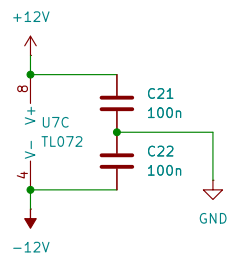
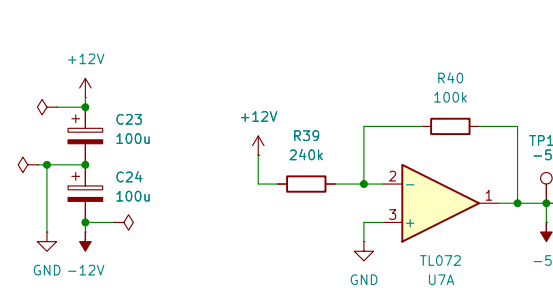
UI power connection



Euro rack PSU connection



Used for VCO current source



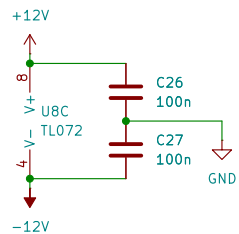
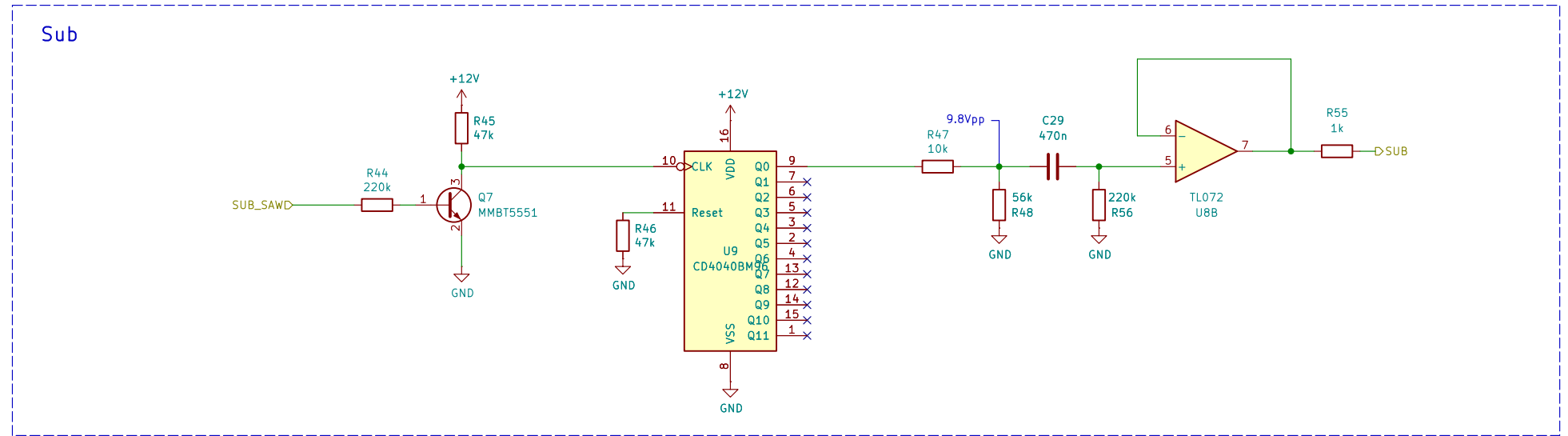
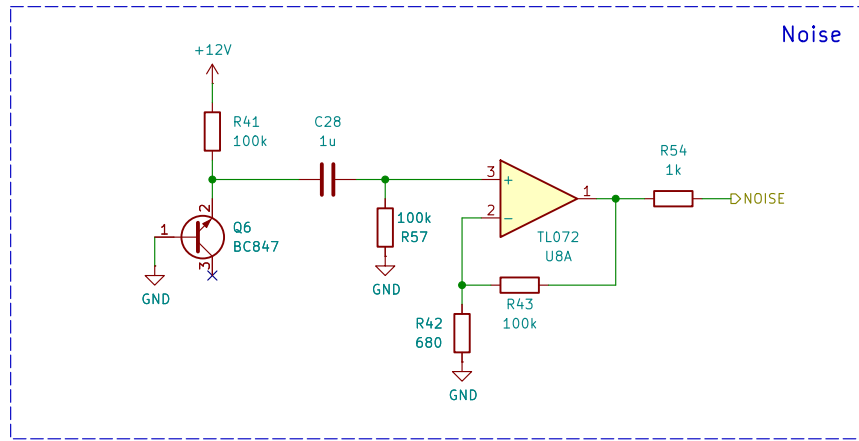
Shmøergh

Sheet: /Power/
File: power.kicad_sch

Title: Moduleur VCO core

Size: A4 Date: 2026-05-24
KiCad E.D.A. 10.0.2

Rev: 3
Id: 4/5



Shmørgh

Sheet: /Sub + Noise/
File: sub+noise.kicad_sch

Title: Moduleur VCO core

Size: A4 Date: 2026-05-24

KiCad E.D.A. 10.0.2

Rev: 3

Id: 5/5